

SPADEADAM NO2, United Kingdom

WMO: 032240

Lat: 55.0503N

Lon: 2.5544W

Elev: 938

StdP: 14.20

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.9	25.4	19.7	15.3	25.2	22.1	17.2	27.8	36.6	43.6	33.3	43.1	6.4	260	0.734

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	72.1	62.0	68.3	59.8	65.2	58.4	63.7	69.5	61.6	66.3	59.8	63.5	10.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.5	84.6	65.6	59.7	79.2	63.2	58.2	74.9	61.3	29.4	69.6	27.9	66.4	26.7	63.6	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.1	26.9	24.0	DB	18.6	78.3	3.4	3.5	16.1	80.8	14.1	82.9	12.2	84.8	9.7	87.4	(4)
			WB	18.0	67.5	3.5	2.8	15.5	69.5	13.4	71.1	11.5	72.7	9.0	74.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	45.5	36.4	36.7	38.9	42.9	48.1	52.9	56.5	56.0	52.2	46.8	41.0	37.1	(6)
(7)		DBStd	8.83	5.16	5.06	5.00	5.01	5.31	4.52	4.32	4.05	4.19	4.92	5.34	6.05	(7)
(8)		HDD50	2297	422	372	344	219	100	20	1	2	23	120	273	401	(8)
(9)		HDD65	7117	887	791	808	663	524	364	265	281	384	564	721	865	(9)
(10)		CDD50	659	0	0	1	6	41	106	204	188	89	22	2	0	(10)
(11)		CDD65	4	0	0	0	0	0	0	2	2	0	0	0	0	(11)
(12)		CDH74	54	0	0	0	0	3	7	26	18	0	0	0	0	(12)
(13)		CDH80	2	0	0	0	0	0	0	1	1	0	0	0	0	(13)
(14)	Wind	WSAvg	11.2	12.8	12.8	12.5	11.0	11.2	10.3	9.5	9.4	10.3	11.3	11.4	11.8	(14)
(15)	Precipitation	PrecAvg	48.0	5.3	4.2	3.1	3.0	3.1	3.7	3.9	3.4	4.7	4.8	5.4	5.4	(15)
(16)		PrecMax	61.0	9.2	10.7	6.0	5.7	5.5	7.8	6.6	8.6	5.6	8.9	12.1	14.3	(16)
(17)		PrecMin	33.7	0.5	0.9	0.9	1.2	0.7	0.7	1.4	0.7	0.4	1.1	2.2	1.3	(17)
(18)		PrecStd	7.4	2.2	2.7	1.2	1.2	1.4	1.7	1.5	1.9	1.3	2.1	2.2	2.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	51.2	56.0	65.1	71.6	73.9	77.7	76.6	69.6	61.1	55.5	52.4	(19)
MCWB			48.9	47.8	47.2	53.4	59.6	62.6	65.0	64.1	60.9	56.0	53.9	51.1	(20)	
2%		DB	47.7	48.4	50.9	57.8	66.1	68.9	72.4	69.5	64.3	58.0	52.6	49.7	(21)	
		MCWB	46.7	46.6	46.7	50.1	55.8	59.5	62.3	62.2	57.9	55.2	51.4	48.5	(22)	
5%		DB	45.9	46.5	48.4	53.9	61.0	64.9	68.6	65.6	61.6	56.1	50.6	47.7	(23)	
		MCWB	44.9	45.1	45.7	48.3	53.4	57.3	60.5	59.6	56.5	54.1	49.5	46.7	(24)	
10%		DB	44.4	44.6	46.5	51.2	57.0	61.5	64.7	62.9	59.2	54.3	48.8	45.8	(25)	
		MCWB	43.4	43.3	44.6	47.0	51.0	55.3	58.8	58.4	55.4	52.4	47.7	44.8	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.3	49.2	50.4	54.7	61.0	64.7	67.0	65.9	62.7	58.2	54.2	51.7	(27)
MCDWB			49.8	49.9	53.8	61.9	69.3	71.0	74.9	73.1	67.0	59.8	54.8	52.0	(28)	
2%		WB	47.0	47.1	48.0	51.4	57.0	61.0	63.9	63.1	59.8	56.2	51.7	49.1	(29)	
		MCDWB	47.6	47.9	49.4	55.7	63.5	67.3	70.0	67.7	62.7	57.3	52.4	49.6	(30)	
5%		WB	45.3	45.3	46.3	49.5	54.5	58.3	61.6	61.2	57.8	54.5	49.7	47.1	(31)	
		MCDWB	45.9	46.3	47.7	52.9	59.6	63.2	66.9	64.4	60.2	55.8	50.4	47.7	(32)	
10%		WB	43.5	43.4	44.7	47.6	52.3	56.3	59.7	59.4	56.1	52.7	47.8	45.0	(33)	
		MCDWB	44.3	44.6	46.3	50.5	56.2	60.3	63.6	61.9	58.4	53.9	48.6	45.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.4	8.7	10.6	13.4	14.7	14.0	13.0	12.3	10.3	8.6	8.1	(35)	
MCDBR			8.9	10.2	13.4	18.3	21.2	20.5	20.4	17.3	15.8	11.7	9.5	9.8	(36)	
MCWBR			8.9	9.0	9.8	11.6	12.6	12.5	11.4	10.6	9.9	9.4	9.0	9.6	(37)	
5% WB		MCDWB	8.9	9.1	11.4	15.8	18.5	18.1	17.7	15.0	13.1	10.7	9.3	9.9	(38)	
		MCWBR	9.0	8.7	9.6	11.0	12.2	12.2	11.2	10.4	9.7	9.6	9.1	9.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.324	0.353	0.374	0.375	0.374	0.374	0.363	0.347	0.332	0.307	0.294	(40)	
(41)		taud	2.273	2.363	2.308	2.279	2.318	2.345	2.347	2.389	2.442	2.440	2.349	2.246	(41)	
(42)		Ebn at Noon	194	240	258	269	275	277	274	270	259	234	196	171	(42)	
(43)		Edh at Noon	18	23	31	37	37	37	36	33	27	22	16	15	(43)	
(44)	All-Sky Solar Radiation	RadAvg	174	380	720	1158	1510	1490	1438	1161	829	451	223	131	(44)	
(45)		RadStd	24	40	72	122	115	147	156	89	70	55	22	12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	(47)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page